

Lesson 10: Project - Make a Library Part 3

45 minutes

Overview

Students complete their library project, finalizing their code and writing written responses explaining the way one of the functions in their library works.

Purpose

This lesson wraps up this quick project but also gives students a chance to practice responding to questions that appear in the Create PT. This lesson and project should therefore serve as a cap on students' study of procedural abstraction and if you are teaching this class as an AP, prepare them for the Create Task that they will complete in the following unit.

Standards

Full Course Alignment

CSP Conceptual Framework

- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming

Agenda

Warm Up

Activity (40 minutes)

Wrap Up (5 minutes)

Assessment

Objectives

Students will be able to:

- Debug a library of functions based on testing and classmate feedback
- Explain the purpose and functionality of a function they developed
- Explain two different calls to a function they developed

Preparation

- Ensure students have access to their project guides
- Review the Example Project Guide and Library in the "Resources" section of the lesson plan
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Bingo Cards - Unit 6 & 7** - Handout [▼ Make a Copy](#)
- **CSP Bingo Generator** - Resource
- **CSP Unit 7 - Lesson 8-11 - Project & Assessment** - Slides [▼ Make a Copy](#)
- **Writing Support - Unit 7 Project - Make a Library** - Handout [▼ Make a Copy](#)

For the students

- **Project Guide - Make a Library** [▼ Make a Copy](#)

Teaching Guide

Warm Up

Remarks

Today is your final day to work on your library. Based on the feedback you received from your classmates and the tests that you wrote, you should know where you need to improve your functions. Today you'll also need to finish a set of written response questions about one of your functions explaining how it works and describing two different calls to that function with different arguments.

Let's get to it!

Activity (40 minutes)

Teaching Tip

Scheduling Time: Depending on where students are in the process of developing their libraries they may need to program a little more prior to starting their free response answers. Remind students that in order to finish their responses they only need to have written a single function that has a parameter, return value, loop, and conditional. As soon as students have completed a function like that they should move to free response answers. They can subsequently come back to finish developing the rest of their libraries.

 **Step 7 - Acknowledge Collaborators / Sources - 2 mins:** If students worked in pairs or imported some code from elsewhere they should acknowledge the sources of that code.

 **Step 8 - Free Response - 20 mins:** With students quickly review the questions students need to answer about one of the functions in their library. These questions are taken nearly exactly from the Create PT and should help students prepare to independently answer these questions when they complete that task.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Code Support: For students who did not fully complete their app, provide them with this adaptation from the exemplar code.

Work Time - 18 mins: Students should work on their free response answers early in the lesson. Once they've finished working on their responses they can go back to their project to make any final edits to their project code. Encourage students to review the Scoring Guidelines to make sure they're not missing anything in their projects.

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Build Your Library

Wrap Up (5 minutes)

 **Submit Projects:** At the end of the lesson students should submit their projects in Code Studio and submit their completed project guides, including free response questions.

CSP Vocabulary Bingo

Remarks

Now that we are near the end of the unit, we are going to play CSP Vocabulary BINGO once again. You have been working hard on learning the definitions of all of the different vocabulary words throughout the unit, and now we will put your amazing memories to the test.

I will give each of you a BINGO card that has randomized vocabulary on it. When you hear the definition for a word that you have on your card, mark that square. If you have 4 in a row horizontally, vertically, or diagonally, yell, "BINGO!"

Are there any questions?

 **Do This:** Pass out a BINGO Card to all students. A printable PDF is located in the Resources section of this lesson plan.

 **Do This:** Use the [**CSP Bingo Generator**](#) for Unit Seven in order to generate clues for the students.

Assessment

Rubric: Evaluate the projects using the provided rubric in the Project Guide

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End of Unit Survey



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